

INFORMATION FOR FIRST AND SECOND RESPONDERS

EMERGENCY RESPONSE GUIDE



Kia Niro PHEV

Li-ion Battery

Li-ion



Introduction

The ERG (Emergency Response Guide) provided by Kia describes emergency response operations, warnings, and precautions related to the vehicle. This publication is intended to provide necessary information for vehicle accident rescue operations and for the training and further education of first and second responders.

Please note that the guide may be updated by Kia on an ongoing basis. It is not intended for retailers, end consumers, or any other readers not mentioned in the preceding sentence.

The provided guide applies only to the Kia Niro PHEV vehicle and includes information about the location and description of high-voltage components and the vehicle's structure. However, it does not cover every scenario in emergency situations.

Failure to follow the recommended procedures during emergency response may result in death or other serious injury. It is important to read the guide in advance as it contains essential information about the vehicle's features and structure in the event of an accident.

IMPORTANT INFORMATION



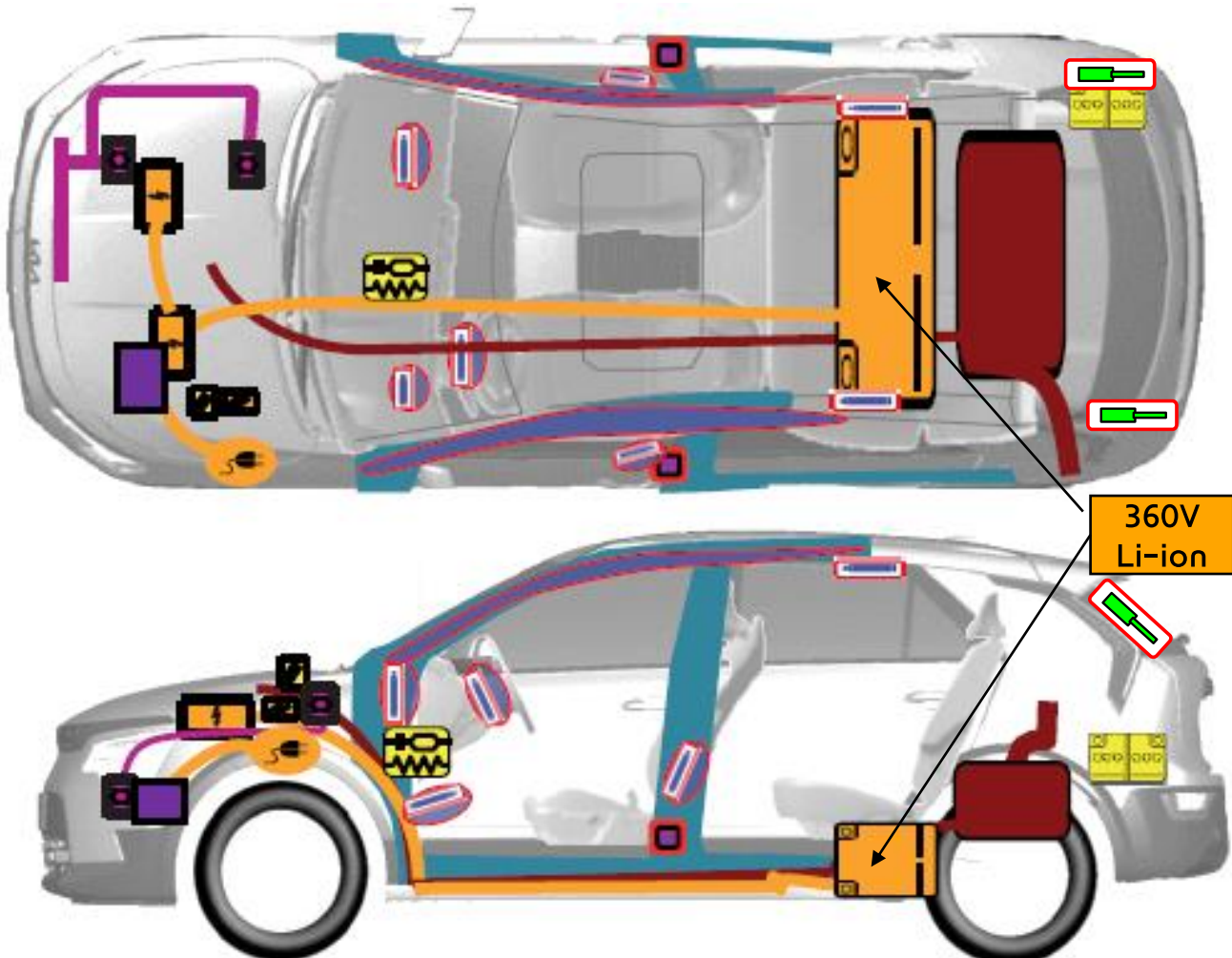
A WARNING indicates a situation in which harm, serious bodily injury, or death could result if the warning is ignored.

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**Kia Niro Plug in Hybrid
SUV, 5doors
From 2023**



	Airbag		Stored gas inflator		Seat belt pretensioner		SRS control unit		High voltage component
	Battery, low-voltage		Gas strut / Preloaded spring		High strength zone		Fuel tank content gasoline / ethanol		Battery pack, high-voltage
	High voltage power cable		Air-conditioning component		Air-conditioning line				

Initial Response: Identify, Immobilize, and Disable

The following procedures should be used whenever you deal with a Niro PHEV at an emergency scene. However, all operations must be consistent with your department's standard operating procedures, guidelines, and applicable laws. When a PHEV is damaged in a crash, the high-voltage safety systems may have been compromised and pose a potential high-voltage electrical shock hazard. Exercise caution and wear appropriate personal protective equipment (PPE), including high-voltage safety gloves and boots. Remove all metallic jewelry, including watches and rings.

WARNING

The Niro PHEV is a hybrid electric vehicle. Emergency responders should properly respond to emergency scenarios involving the Niro PHEV and exercise extreme care to avoid contact with the high-voltage system within the vehicle.

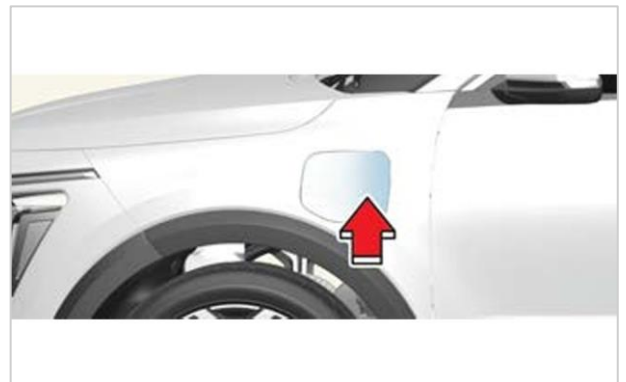
Charging Door

The charging door is located on the side of the vehicle.

Note that the doors must be unlocked to open the charge port door.

How to open the charging door

1. Depress the brake pedal and apply the parking brake.
2. Turn OFF all switches, move the shift lever or dial to P (Park), and turn OFF the vehicle.
3. Open the charging door by pressing rear center edge of the charging door.
4. Remove any dust on the charging connector and charging inlet.



Unlock charging door in emergency



If the charging connector does not disconnect due to battery discharge and/or failure of the electric system, open the luggage board, remove the storage box and lightly pull the emergency cable. The charging connector will then disconnect.

Identifying a Niro PHEV

Vehicle Identification Number (VIN)

The Vehicle Identification Number (VIN) is the number used in registering your vehicle and in all legal matters pertaining to its ownership, etc. The number is punched on the floor under the front right side seat. To check the number, open the cover.

Type A



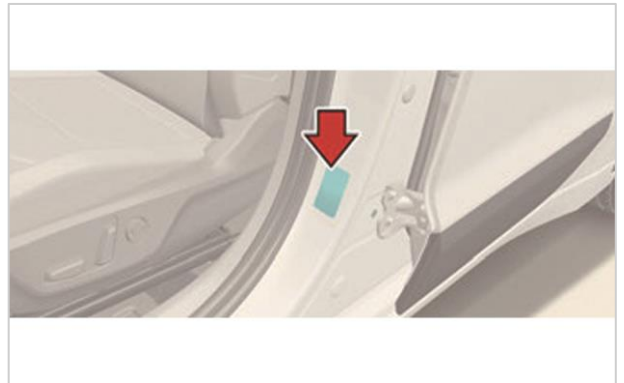
The VIN is also on a plate attached to the top of the dashboard. The number on the plate can easily be seen through the windshield from outside.

Type B



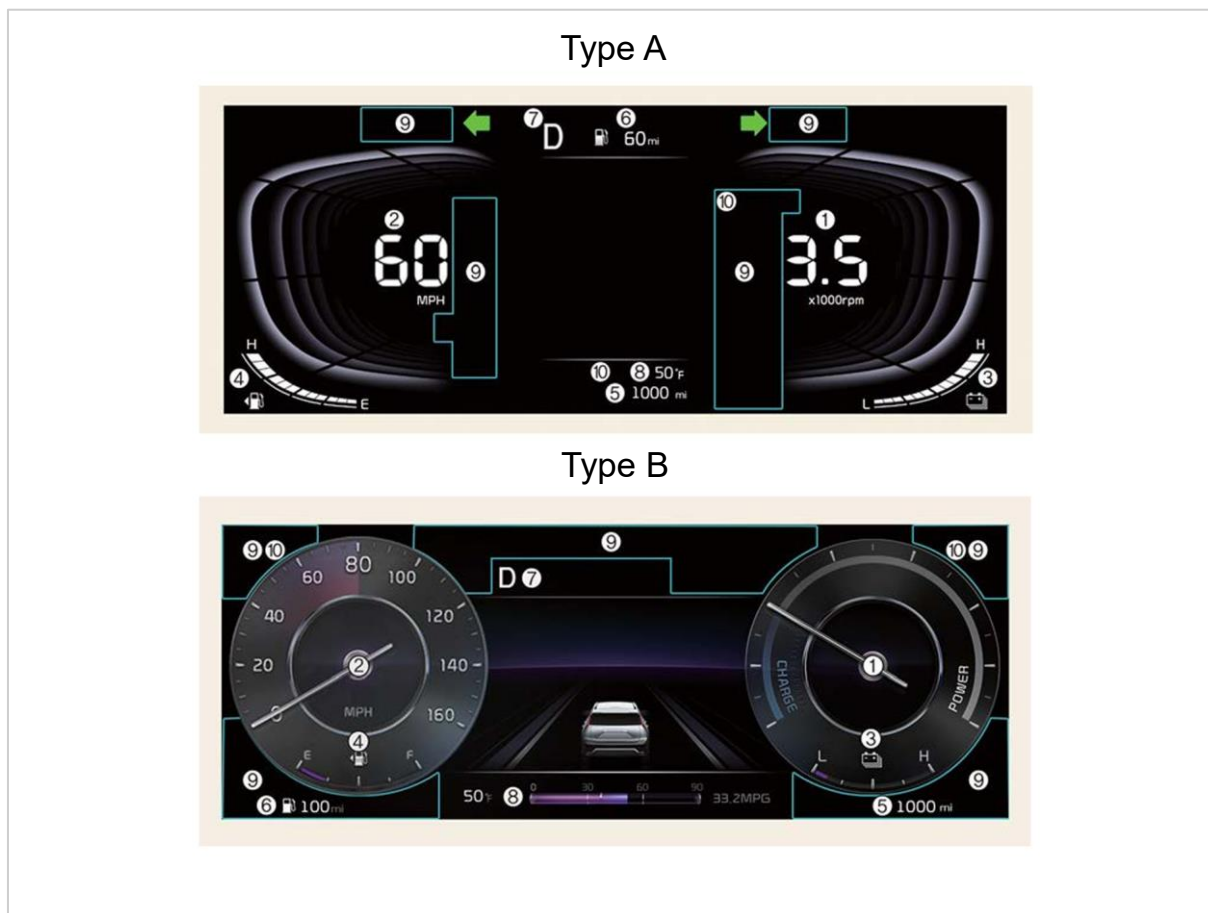
Vehicle certification label

The vehicle certification label attached on the driver's side center pillar gives the vehicle identification number (VIN).



Niro PHEV Cluster Instrument Panel

The Niro PHEV Cluster Instrument Panel displays the PHEV specific features such as battery pack, high-voltage SOC (State of Charge) as below.



1. Tachometer / Power gauge
2. Speedometer
3. Hybrid battery SOC gauge
4. Fuel gauge
5. Odometer
6. Distance to empty
7. Transmission shift indicator
8. Outside temperature gauge
9. Warning and indicator lights
10. Plug-in hybrid mode indicator

Charge Status Indicator

You can check the charging status at the outside of vehicle when charging or using(it is not driving status) the battery pack, high-voltage. It may take about 2-3 hours or more to complete charging, depending on conditions.



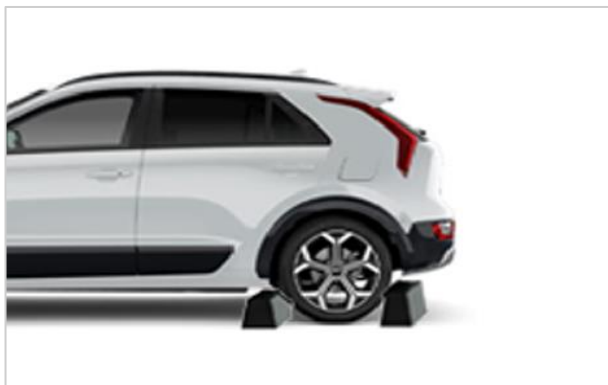
Classification		Details		Operation of charging indicator lamp	
Mode	Status			Charging inlet	Charging button
A	READY	Non-charging state	0 ~ 65 %	On (Yellow)	Off
			65 ~ 100 %	On (Green)	Off
B	Aux. Battery Saver/High voltage warning	Charging the 12V auxiliary battery / High voltage warning state		On (Red)	Off
C	Charging	Charging	0 ~ 65 %	Blinking (Yellow)	Off
			65 ~ 100 %	Blinking (Green)	Off
D	Charging complete	Charging completed (turns off in 5 seconds)	100 %	On (Green) > OFF	Off
E	Charging failed	Error while charging		Blinking (Red)	Off
F	Scheduled charging standby	Reserved charging is operating (turns OFF after 3 minutes)		Off	On
G	Error	CAN communication error status		On (Green) > On (Orange) > On (Red)	Off

2.1 Immobilization

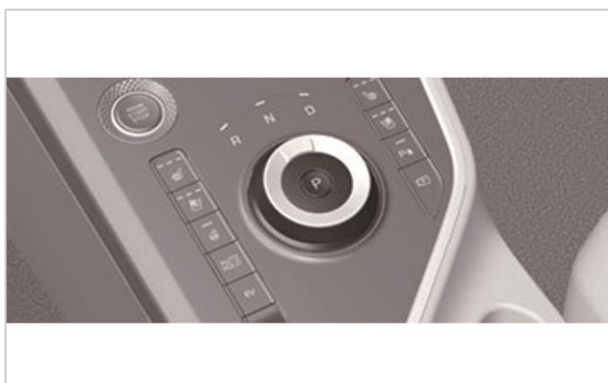
The next step is to immobilize the vehicle to prevent any accidental movement that can endanger response personnel or civilians. When the Niro PHEV is damaged in a crash, the vehicle may appear to be shut off, even when it actually is not, due to a lack of virtual engine sounds.

When the "READY" mode light is illuminated on the Instrument Panel, the vehicle can move silently using the electric motor. Responders should approach the vehicle from the sides and stay away from the front or rear as they are potential paths for vehicle movement. Be sure to immobilize the vehicle in the following manner.

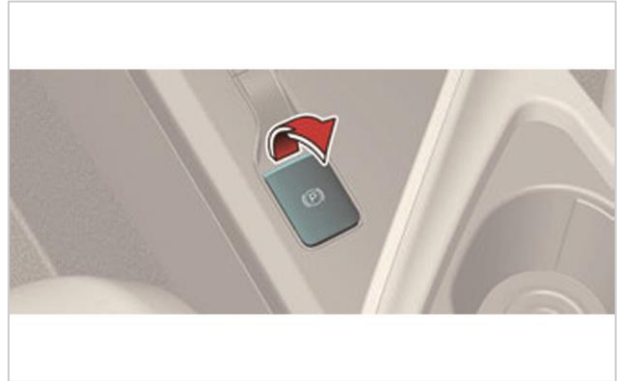
1. Chock the wheels.



2. Put the vehicle in P (Park) position by pressing the 'P' button.



3. Set the Electronic Parking Brake (EPB)



WARNING

- When installing a block or jack, avoid the high voltage power cable, the battery pack, high-voltage, and the fuel system (exhaust system).
- If high-voltage components or cables are exposed, do not place any support on them.

2.2 Vehicle Stabilization

Use the standard stabilization (lift) points, as shown below. Always check the connection to the structural component of the vehicle and avoid placing cribbing under the high-voltage power cables and other areas not normally considered acceptable.



Disabling vehicle

The final step in the initial response process, after immobilizing the vehicle, is to disable the vehicle, its SRS components, and the high-voltage electrical system. To prevent current from flowing through the system, use one of the following procedures to disable the vehicle.

Case 1:

1. Engine – OFF
2. Battery, low-voltage disconnection
3. High-voltage shut off

Case 2*:

1. Fuse removal (Fuse box)
2. Battery, low-voltage disconnection
3. High-voltage shut off

*In case of emergency

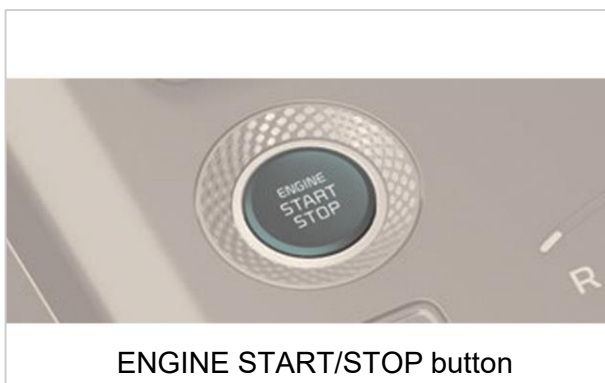
SRS : Supplemental Restraint System

3.1 Disabling the System – Smart Key System and “ENGINE” START/STOP Button

1. Confirm the status of the READY light on the instrument panel. If the READY light is on, the vehicle is ON.

a) If the READY light is NOT on, the vehicle is off. Do not push the “**ENGINE**” **START/STOP** button because the vehicle may start (go into READY mode).

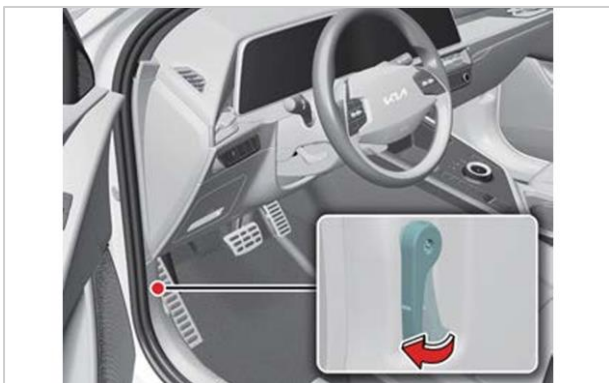
b) To turn OFF the system, shift to the 'P' (Park) position on the shifter dial and press the ENGINE START/STOP button.



2. If necessary, lower the windows, unlock the doors, and open the liftgate as required before disconnecting the battery, low-voltage. Once the battery, low-voltage is disconnected, power controls will not operate.

3. Before disconnecting the battery, low-voltage, remove the smart key at least 2 meters (7 feet) away from the vehicle to prevent accidental restart.

4. Pull the release lever to unlatch the hood. The hood should pop open slightly.



5. Go to the front of the vehicle, raise the hood slightly, push the secondary hood release lever (1) to the left and lift the hood (2).

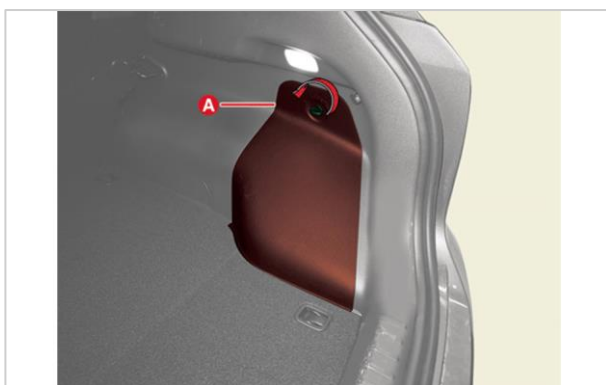


6. Follow the procedures below to remove the service interlock connector and disable the battery pack, high-voltage:

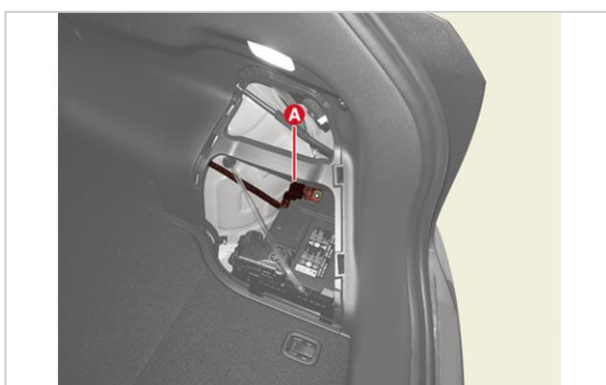
Disable the battery pack, high-voltage

1. Disconnect the battery negative (-) terminal and the service interlock connector.

(1) Remove the upper battery cover (A) after rotating valve in the direction of the arrow.

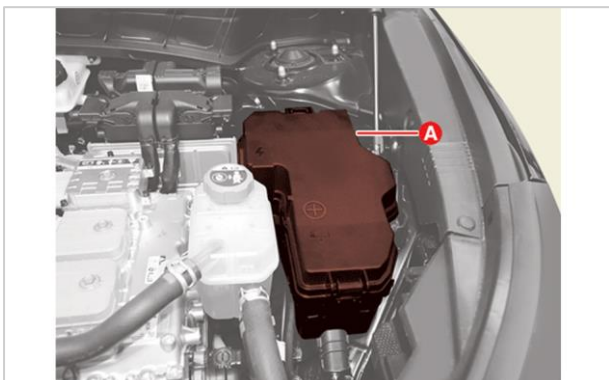


(2) Disconnect the negative connector from the auxiliary battery, low-voltage (A) after loosening the nut.



3. DISABLE DIRECT HAZARDS / SAFETY REGULATIONS

(3) Remove the engine room junction block cover (A).



(4) Disconnect the service interlock connector (A).

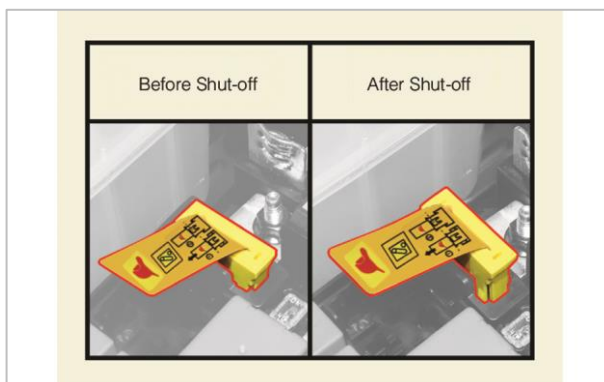


WARNING

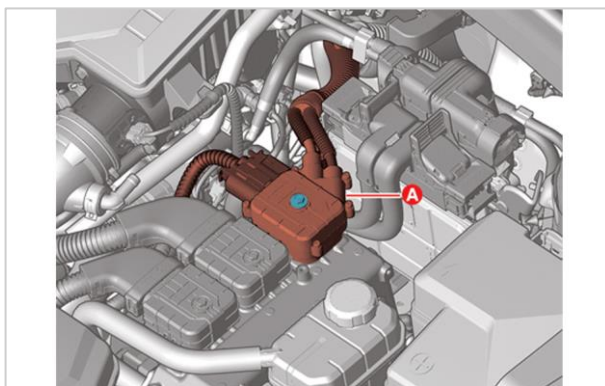
Wait more than 3 minutes for capacitor in the inverter to be discharged completely after disconnecting the high voltage service interlock connector.

NOTE

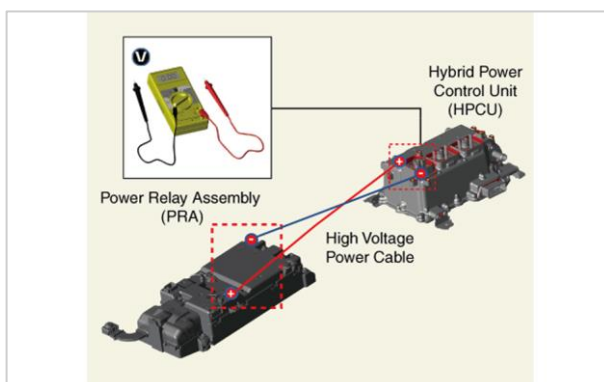
The high voltage service interlock connector cannot be completely removed.



2. Disconnect the inverter power cable (A) from the Hybrid Power Control Unit (HPCU) after loosening the bolt.

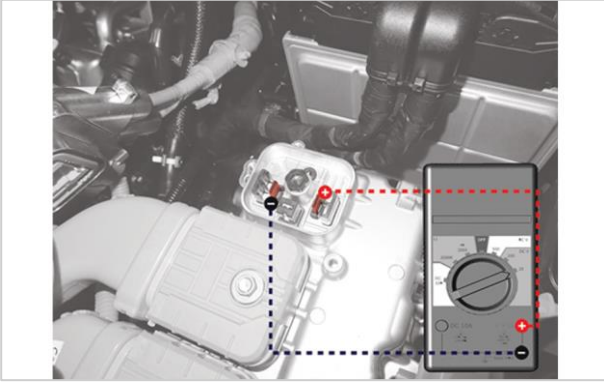


3. Measure the voltage between the inverter (+) terminal and the inverter (-) terminal.

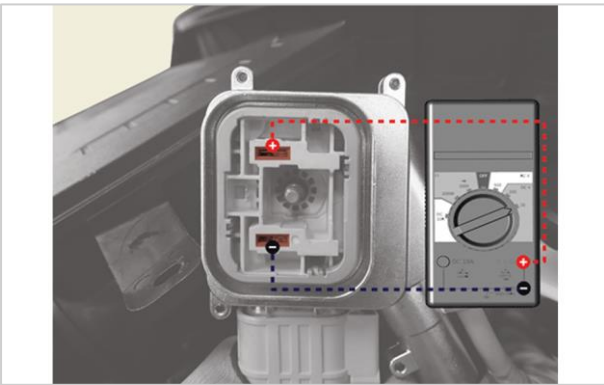


3. DISABLE DIRECT HAZARDS / SAFETY REGULATIONS

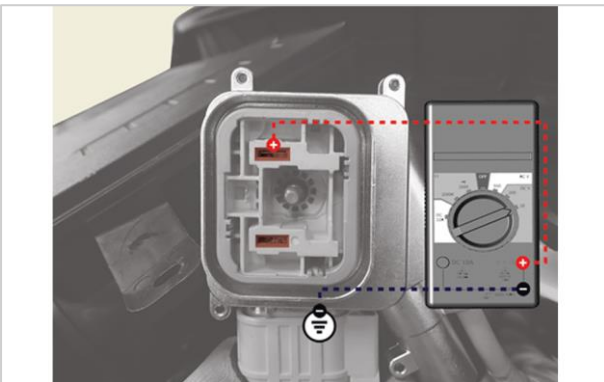
(1) Measure the voltage between the inverter (+) terminal and the inverter (-) terminal.



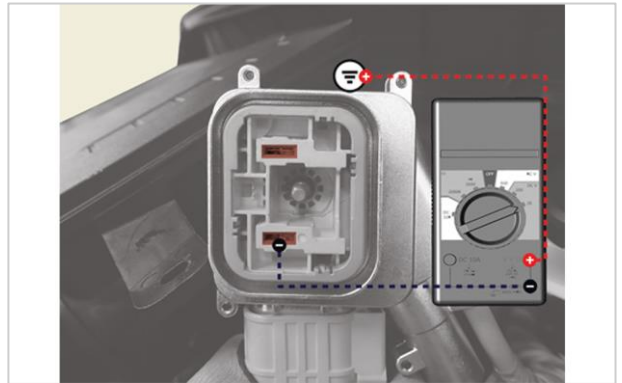
(2) Measure the voltage between the power cable (+) and (-) terminal.



(3) Measure the voltage between the power cable (+) terminal and the ground.



(4) Measure the voltage between the power cable (-) terminal and the ground.

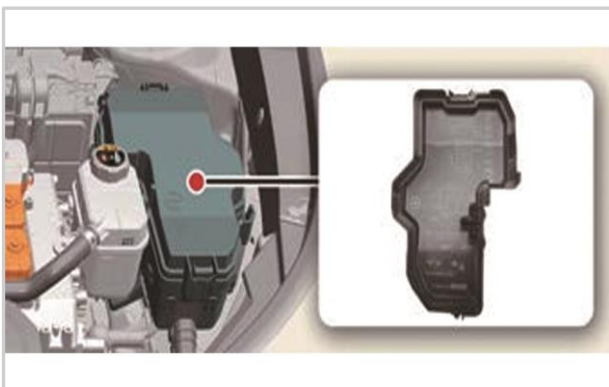


WARNING

If the voltage of 30V or higher is measured, check if the high voltage service interlock connector is removed. If the voltage of 30V or higher is measured even though the high voltage service interlock connector is removed, there may be a significant problem with the high voltage circuit. In this case, perform DTC and do not touch the parts related to the high voltage system.

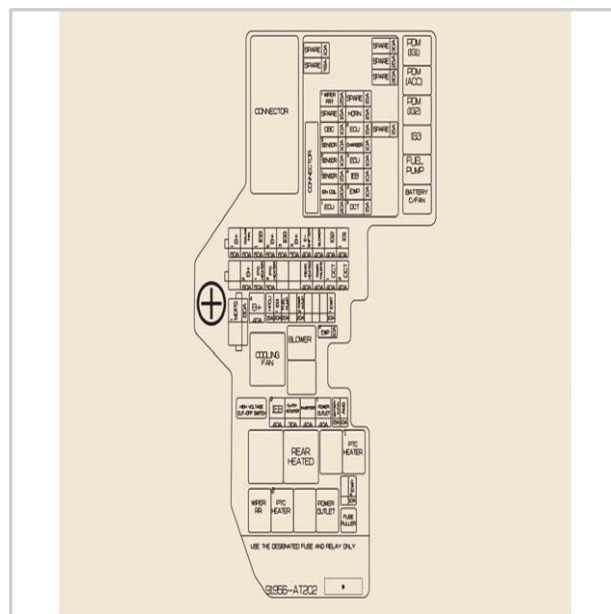
3.2 Disabling the system – IG (Ignition) Fuse Removal

1. Open the hood.
2. Remove the engine room fuse box cover.
3. If necessary, lower the windows, unlock the doors and open the liftgate as required, before disconnecting the battery, low-voltage. Once the battery, low-voltage is disconnected, power controls will not operate.
4. In the event the vehicle is unable to be disabled using the "ENGINE" START/STOP Button, use the fuse puller located in the engine room fuse box and pull both the IG1 and IG2 fuse from the engine room fuse box. If the IG fuses cannot be found, pull out all of the fuses and relays in the fuse box.



Engine room fuse box

- The actual engine room in the vehicle may differ from the picture.



5. Remove the service interlock connector and disable the battery pack, high-voltage.

If previously mentioned methods of disabling the vehicle's system are unsuccessful, any emergency procedures involving the PHEV vehicle may cause the accidental deployment of undeployed airbags and electric shock from high-voltage components.



WARNING

Electrocution risk

- Before engaging in any emergency response procedures, ensure the vehicle is disabled and wait 5 minutes to allow the capacitor in the high-voltage system to be discharged to avoid electrocution.
- Exposed cables or wires may be visible inside or outside the vehicle. Never touch the metal chassis wires, cables, connectors, or any electric components before disabling the system, and/or being shorted to the vehicle chassis.

Failure to follow these instructions will lead to serious bodily injury or death by electrocution.

4.1 Extraction Operations

The Niro PHEV is an eco-friendly vehicle. Because of the high-voltage components within, the first responders should pay special attention when they extract occupants from the vehicle. Before performing extraction operations, the first responders should “Identify, Immobilize, and Disable” the vehicle as explained in sections on emergency procedures.

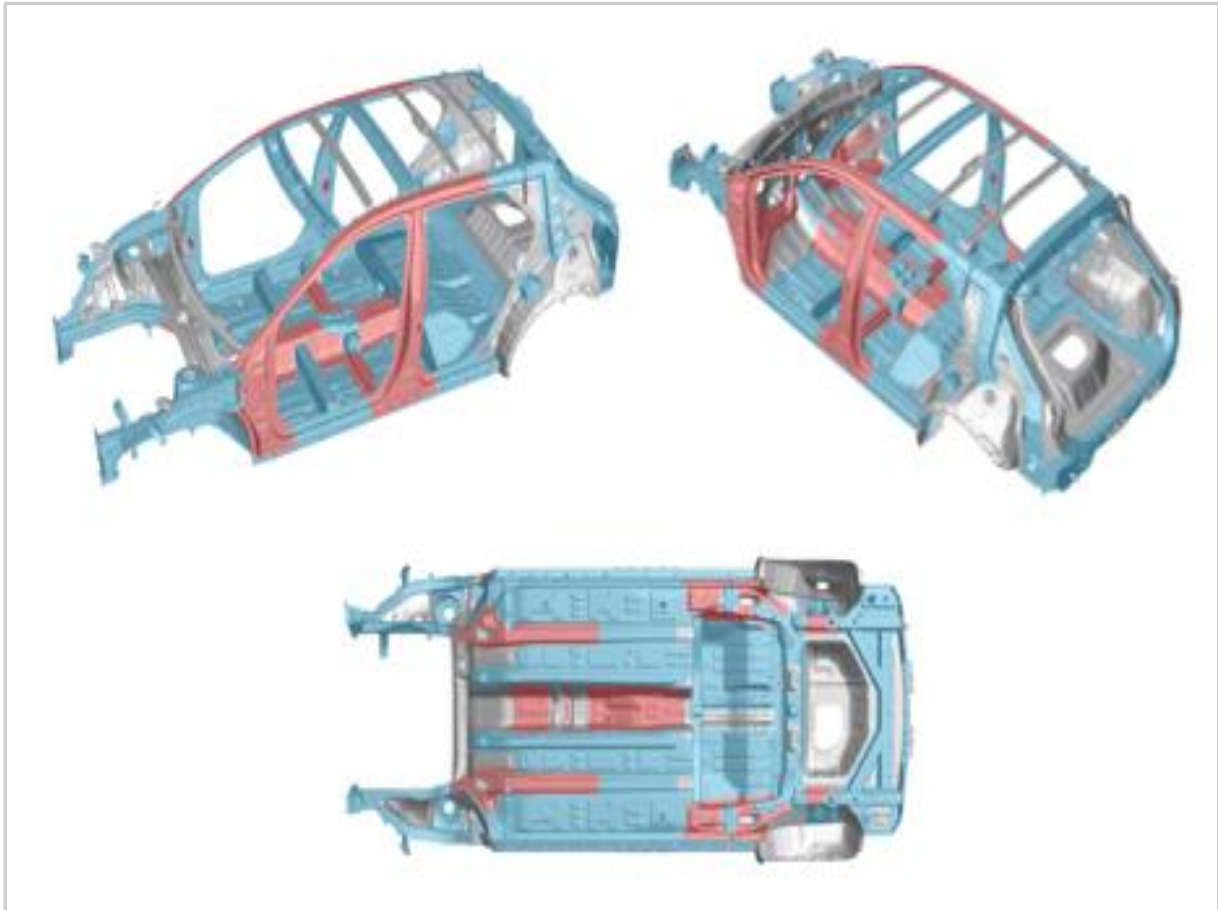
4.2 Extraction Tools and Procedures

When responding to an incident involving the Niro PHEV, we recommend that the first responders follow their organization’s standard operating procedures for dealing with vehicle emergencies.

When the first responders cut the vehicle, they should always pay special attention to the airbag system, orange-colored high voltage power cables, and other high-voltage components so that the parts are not damaged and prevent a risk of explosion or electrocution.

4.3 Location of High Strength Zone

In these images, advanced high strength zone is used in the areas colored in blue and high strength zone is used in the red colored areas. Depending on the tools used, high strength zone can be challenging or impossible to cut. If necessary, use a workaround technique.

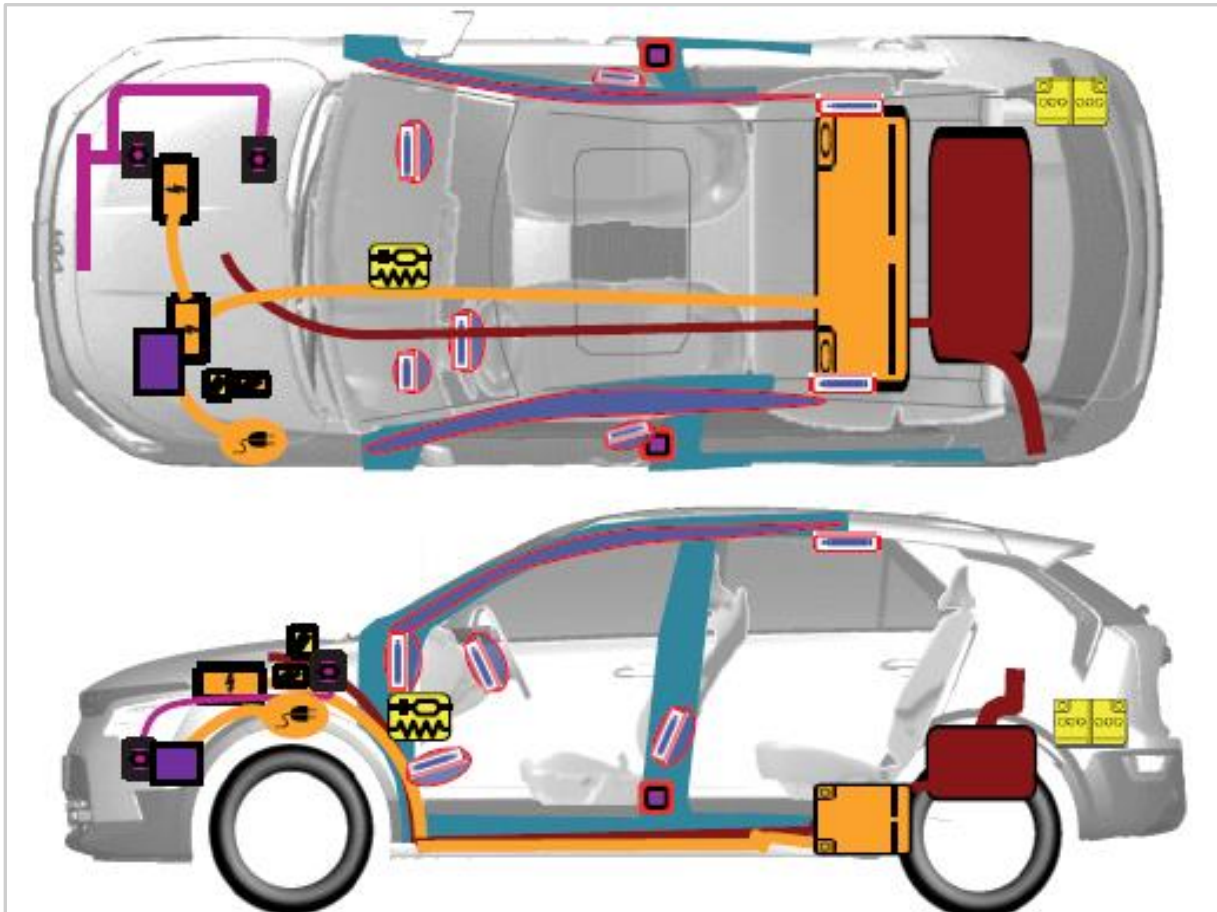


-  Mild zone
-  Advanced high strength zone
-  High strength zone

4. ACCESS TO THE OCCUPANTS

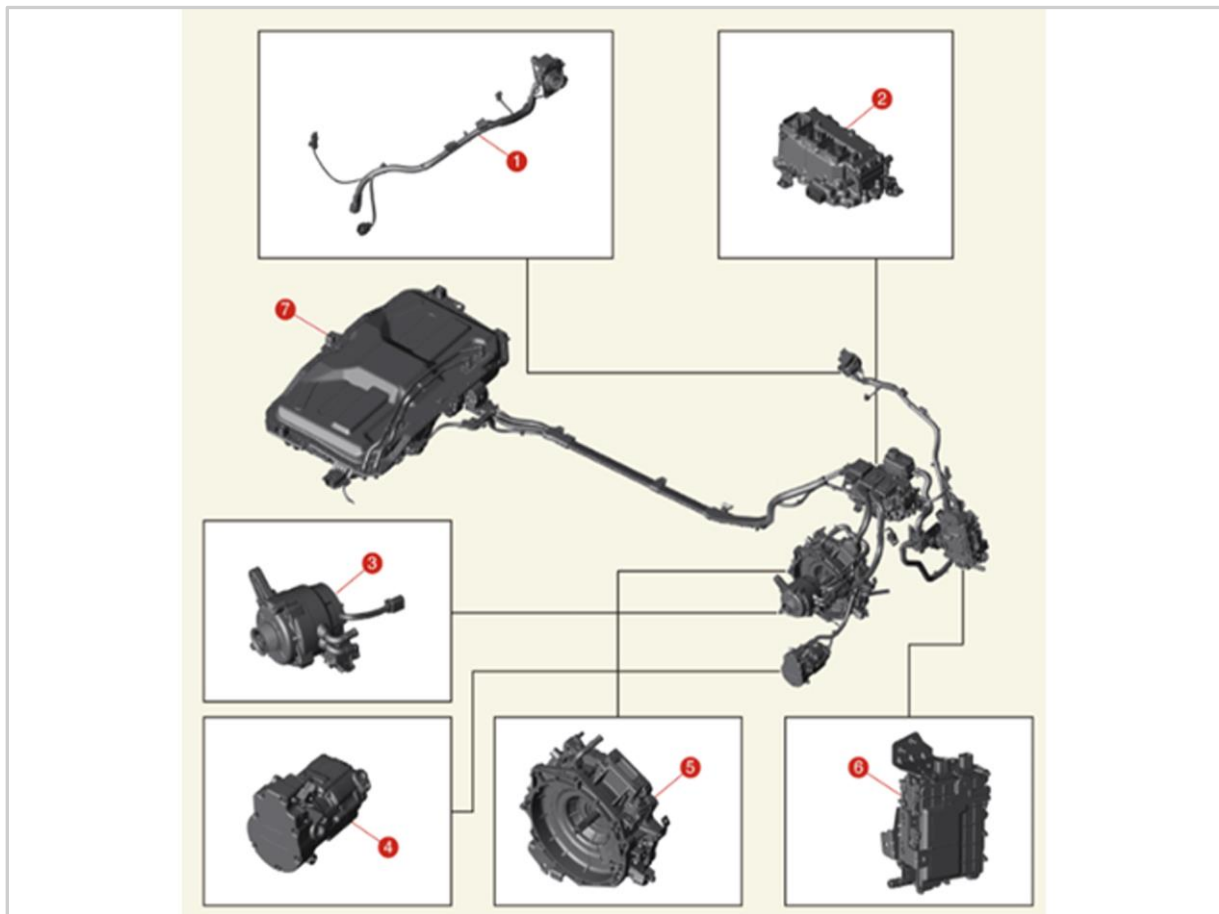
4.4 Occupants Rescue Guide

When dealing with an emergency, refer to the components as below.
Do not cut the body near the airbag, high voltage power cable, and fuel system.



 SRS control unit	 Airbag	 Stored gas inflator
 High-voltage power cable	 Seat belt pretensioner	 Gas strut / Preloaded spring
 Battery pack, high-voltage	 High-voltage component	 High strength zone
 Battery, low-voltage	 Air-conditioning component	 Air-conditioning line
 Fuel tank content gasoline / ethanol		

5.1 Hybrid Control System



- | | |
|-------------------------------------|----------------------------------|
| 1. Normal Charge Port | 5. Hybrid Motor |
| 2. Hybrid Power Control Unit (HPCU) | 6. On-Board Charger (OBC) |
| 3. Hybrid Starter Generator (HSG) | 7. Battery System Assembly (BSA) |
| 4. Electric A/C Compressor | |

Niro PHEV Specification

Capacity	30.8 Ah
Voltage	360 V
Energy	11.1 kWh

5.1 High-voltage System

Hybrid Drive Motor Assembly

The PHEV motor converts electrical energy into motive force with a Max. power of 62 kW and Max. torque of 203 Nm.



HPCU (Hybrid Power Control Unit)

The HPCU includes an Inverter and LDC (Low Power DC-DC Converter) in one housing. The inverter converts DC to AC to supply electricity to the motor. It also converts AC to DC to charge the battery pack, high-voltage. The LDC transfers high voltage electricity to 12 voltage to charge auxiliary battery, low voltage.



Battery Pack, High-Voltage

Battery pack, high-voltage modules supply and store electric energy from traction motor and it is a Lithium ion battery with specifications, 360 V / 30.8 Ah / 11.1 kWh.



HSG (Hybrid Starter Generator)

The HSG restarts the engine in the HEV mode and also charges the battery pack, high-voltage while driving, as this is the generator for hybrid vehicle.



On-Board Charger (OBC)

The OBC is the battery charging equipment that converts external AC to DC to charge the battery pack, high-voltage.



5.2 High-voltage Orange Cabling

The high-voltage cabling is orange, as per Society of Automotive Engineers (SAE) standards.

Cables run under the floor of the vehicle and connect the battery pack, high-voltage to the HPCU, Motor, LDC, Inverter, A/C compressor, and other High-Voltage components located towards the front of the vehicle.

You can identify the vehicle as a PHEV with the presence of orange cables under the hood, in the under-floor battery compartment, or HV cables under the car.



WARNING

High Voltage Power Cables

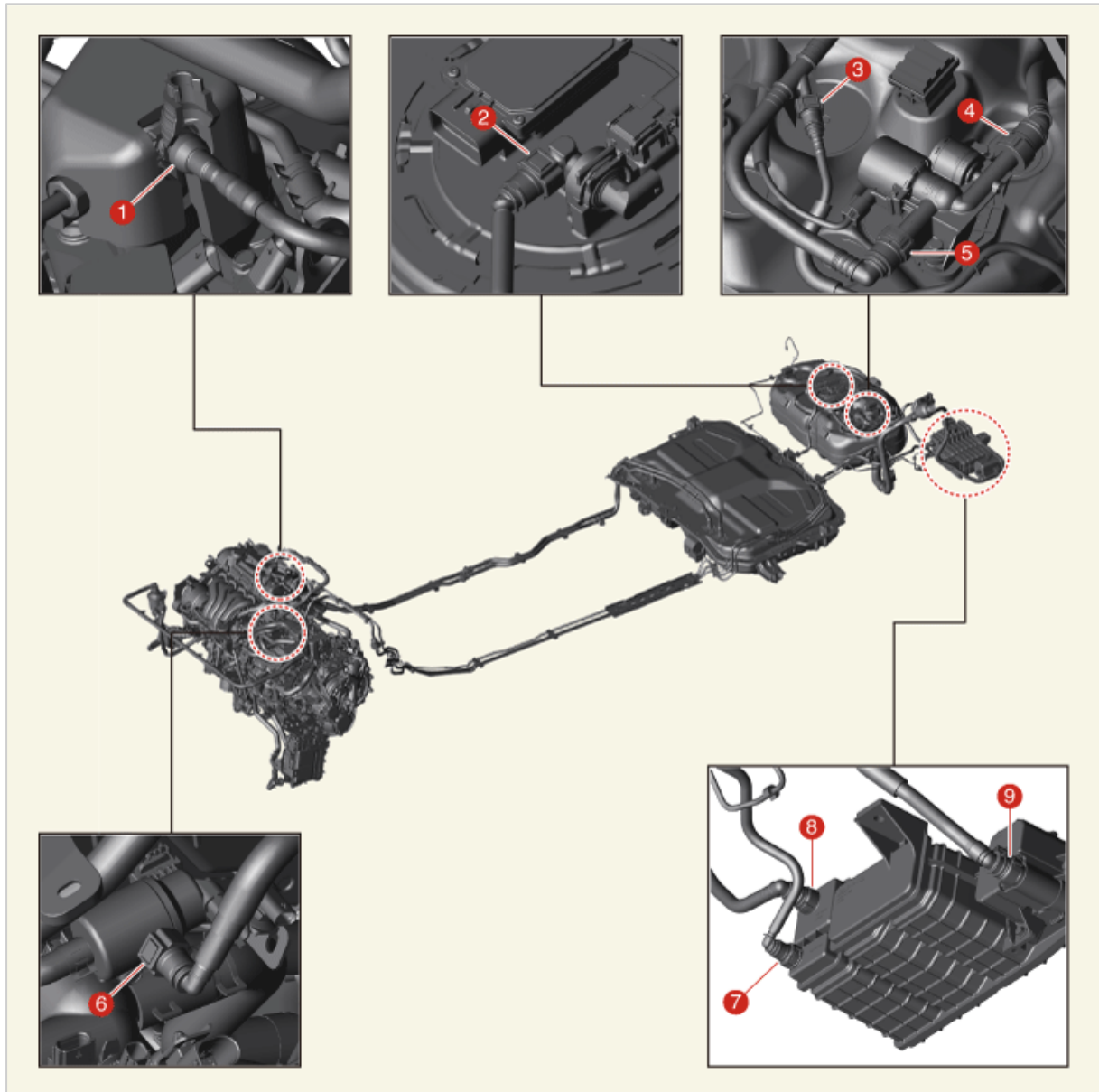
- Never cut or disconnect the high-voltage orange cabling and connectors without first disabling the HV system by removing the service interlock connector. (refer to page 11-13)
- Exposed cables or wires may be visible inside or outside the vehicle. Never touch the metal chassis wires, cables, connectors, or any electric components before disabling the system, or if they are shorted to the vehicle chassis.

Failure to follow these instructions will lead to serious bodily injury or death by electrical shock.



5.3 Fuel (Gasoline)

Niro PHEV has a 1.6L 4 cylinder engine. The fuel type is gasoline, which is stored in the fuel tank content gasoline / ethanol and delivered through the fuel line. When dealing with an emergency, be careful not to cut the fuel line and tank.



1. Fuel Feed Tube (High Pressure Fuel Pump Side)
2. Fuel Feed Tube (Fuel Tank Content Gasoline / Ethanol Side)
3. Vapor Tube (Fuel Tank Content Gasoline / Ethanol ↔ Filler-neck)
4. Vapor Tube (Fuel Tank Isolation Valve ↔ Canister)
5. Vapor Tube (Fuel Tank Content Gasoline / Ethanol ↔ Fuel Tank Isolation Valve)
6. Vapor Tube [Purge Control Solenoid Valve (PCSV) Side]
7. Ventilation Tube (Canister Side)
8. Vapor Tube (Fuel Tank Isolation Valve ↔ Canister)
9. Ventilation Tube (Canister ↔ Fuel Tank Air Filter)

6.1 Firefighting Operations

Strict precautions must be taken while conducting firefighting operations following Reasons:

- Lithium-ion batteries contain liquid electrolytes that can vent, ignite, and spark when it is exposed to the temperature above 300°F (150°C).
- Vehicle may burn rapidly with a flare-burning effect.
- Even after the battery pack, high-voltage fire appears to have been extinguished, renewed or delayed fire can occur.
- Use a thermal infrared camera to ensure the battery pack, high-voltage is completely cooled before leaving the incident.
- Always inform the second responders that there is a risk of the battery re-igniting.
- In a fire, submersion, or a collision that damages the battery pack, high-voltage, always store it in an open area with no exposures within 50 feet (15m).
- A burning battery could release hydrogen fluoride, carbon monoxide, and carbon dioxide gasses. Use full-face self-contained breathing apparatus (SCBA) approved by NIOSH/MSHA with full protective gear. Even if the battery pack, high-voltage is not directly involved in a vehicle fire, approach the vehicle very carefully.

NIOSH: National Institute of Occupational Safety & Health
MSHA: Mine Safety and Health Administration

6.2 Extinguishers

- Small fires not involved with the battery pack, high-voltage should be extinguished with an ABC fire extinguisher. (ex. A fire caused by wiring harnesses or electrical components, etc.)
- Do not attempt to extinguish fires that involve the battery pack, high-voltage with a small amount of water as this can result in electrocution. Fires that involve the battery pack, high-voltage should be extinguished using a large amount of water(Max 100,000 liter) to cool the battery pack, high-voltage. Firefighters should not hesitate to pour a larger amount of water on the vehicle in such scenarios. Make sure the battery is fully cooled to avoid fire re-ignition.

6.3 How to Deal with the Situation

When there is fire:

Extinguish the fire with a large amount of water. Don't use seawater or salt water. It can generate the toxic vapors or cause the re-ignition.

When damaged battery or fluid leak:

- Disconnect battery, low-voltage (-) terminal.
- Disable the high-voltage system.
- Neutralize the battery by applying a large volume of water. (Note: this doesn't discharge the battery)
- Discharge the battery.

*If electrolyte solution leakage, or any damage to the battery pack, high-voltage casing is observed.

6.3.1 Vehicle Fire

- Use a large volume of water (max. 100,000 liters). Water must cool down the battery.
- Adding water to the battery pack, high-voltage casing can enhance the battery's cooling process. However, it is crucial to avoid attempting to penetrate the HV battery or its casing in order to apply water.
- Introducing water into the battery pack, high-voltage can be difficult due to the battery case.
- Introduce water into any openings that may have resulted from the accident or fire.



Battery pack, high-voltage

6.3.2 Battery pack, high-voltage Damage and Fluid Leaks

If electrolyte solution leakage, or any damage to the Lithium ion battery casing is observed, the first responders should wear appropriate Personal Protective Equipment (PPE) and attempt to neutralize the battery by applying a large volume of water to the battery pack. The neutralization process helps stabilize the thermal condition of the battery pack but does not discharge the battery.

- Do not put any smoke, spark, flame around the vehicle.
- Do not touch or step on the spilled electrolyte.
- If electrolyte leak occurs, wear appropriate solvent resistant PPE and use soil, sand, or a dry cloth to clean up the spilled electrolyte. Be sure to adequately ventilate the area.

WARNING

Electrolyte Irritation

The battery pack, high-voltage contains electrolytes. To avoid exposure to electrolyte solution and serious personal injury, always wear appropriate solvent resistant PPE (Personal Protective Equipment) and SCBA (Self-Contained Breathing Apparatus).

- Electrolyte solution is an irritant to the eyes. In the event of contact with eyes, rinse with plenty of water for 15 minutes.
- Electrolyte solution is a skin irritant. Therefore, in the event of contact with skin, wash off with soap.
- Electrolyte liquid or fumes contacting with water will create vapors in the air by oxidization. These vapors may irritate the skin and eyes. In the event of contact with vapors, rinse with plenty of water and consult a doctor immediately.
- Electrolyte fumes (when inhaled) can cause respiratory irritation and acute intoxication. Inhale fresh air and wash your mouth with water. Consult a doctor immediately.

6.4 Battery pack, high-voltage Re-ignition by Stranded Energy

Damaged cells in the battery pack, high-voltage can cause thermal run away* and re-ignition. To prevent re-ignition, the first responder and second responders need to be aware of the risk of stranded energy* which remains in the damaged cells and leads to re-ignition.

*Thermal runaway: The cause of thermal runaway is generally short circuiting inside a battery cell and a resulting increase in the cell's internal temperature. Battery produces heat with thermal runaway and it can spread from one battery cell to other cells, in a domino effect.

*Stranded energy: Energy remains inside any undamaged battery cells after the accident. That stranded energy can cause a battery pack, high-voltage to re-ignite multiple times after a fire has been extinguished.

How to Prevent Re-ignition (Mitigating Stranded Energy Risk)

1. Battery, low-voltage (-) terminal disconnection (To depower battery management system)
2. High-voltage shut off (refer to page 11-13)
3. Discharging the battery pack, high-voltage (refer to page 29)

7.1 Submerged or Partially Submerged Vehicles

Some emergency responses can involve a submerged vehicle. A Niro PHEV that is submerged does not have high-voltage components on the vehicle's body or framework. It is safe to touch the vehicle's body or framework if there is no severe damage to the vehicle, whether it is in water or on land. In the event that the vehicle is submerged or partially submerged, take out the vehicle from the water before attempting to disable the vehicle. Drain the water from the vehicle. Use one of the methods described in page 11-13 to disable the vehicle. Then, discharge the battery by referring to page 29.



WARNING

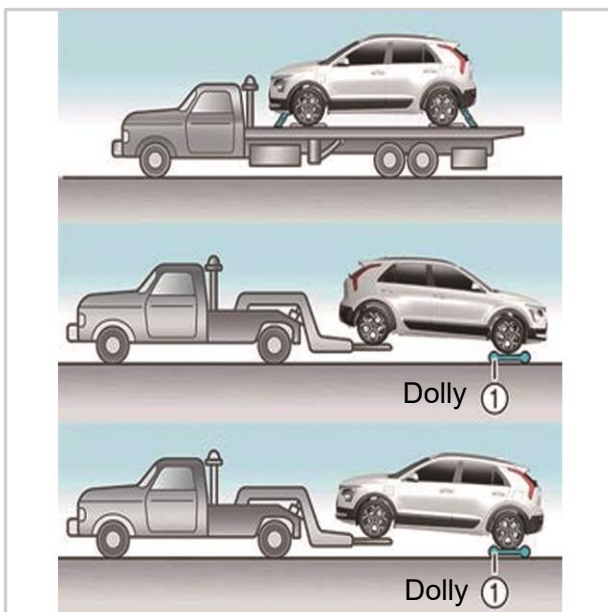
- If severe damage causes high-voltage components to become exposed, responders should take appropriate precautions and wear appropriate insulated personal protective equipment.
- Do not attempt to remove a service interlock connector while the vehicle is in water.

Failure to follow these instructions can lead to death or serious injury by electrocution.

8.1 Towing and Transportation

In the event of an accident, the high-voltage system must be disabled. The service interlock connector must be removed from the battery pack, high-voltage according to one of the methods described in page 11-13 to disable the vehicle. Towing the Niro PHEV vehicle is not different from towing a conventional AWD vehicle.

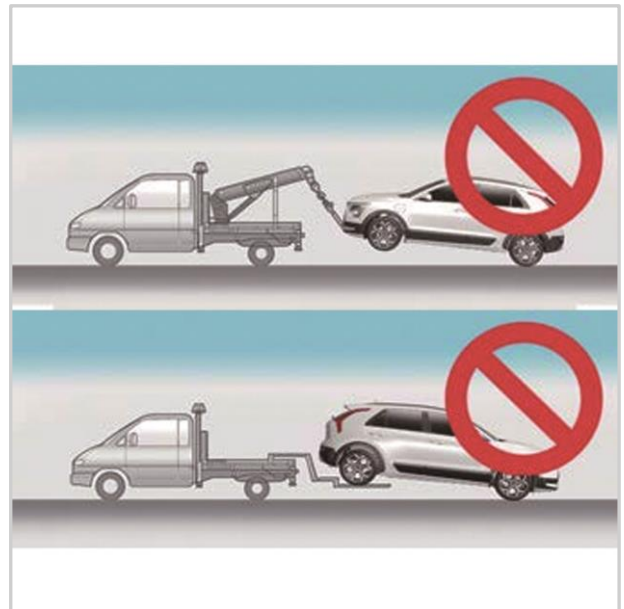
If emergency towing is necessary, we recommend having it done by an authorized Kia dealer or a commercial tow-truck service. Proper lifting and towing procedures are necessary to prevent damage to the vehicle. The use of wheel dollies or flatbed is recommended.



WARNING

- Do not tow the vehicle with the driven wheels of 2WD and front/rear wheels of AWD on the ground as this may cause damage to the vehicle (the vehicle motor may generate electricity and the motor components may be damaged or a fire may occur.)

- Do not tow with sling-type equipment. Use wheel lift or flatbed equipment.



8.2 Storage of damaged vehicle with the damaged battery

- Drain fluids and water, then disconnect the negative (-) terminal of the battery, low-voltage before storing a damaged vehicle.
- In addition, remove the water inside the battery or vehicle, then remove the service interlock connector from the battery pack, high-voltage before storing a damaged vehicle.
- Place the vehicle in an open space away from any structure, vehicle, or building.
- Then, monitor the vehicle until the discharging procedures are completed.
- If the battery can be removed from the vehicle by moving the vehicle on the lift, remove and discharge the battery. (refer to the next page)

- In the event that the battery cannot be taken out, set a water pool and pour water until the entire battery is immersed.
- # Water pool condition: tap water or pond water that does not contain salt
- Maintain this water level for at least 90 hours.
- Then, put salt into the water pool to make 3.5 % salt water.
- Wait for additional 48 hours in salt water.
- Drain the water and dry it.
- If it is impossible to remove or discharge the battery from the vehicle or submerge the vehicle, put a waterproof cover on the vehicle. (prevention of rainwater)
- Even after the battery pack, high-voltage fire appears to have been extinguished, renewed or delayed fire can occur.
- In a fire, submersion or a collision that damages the battery pack, high-voltage, always store it in an open area with no exposures within 50 feet (15m).

WARNING

Battery discharging

- DO NOT USE SALT WATER for the first step.
- A large volume of flammable hydrogen gas can be generated in salt water due to electrolysis.
- After submerging the vehicle in pure water for at least 90 hours, put salt in the water pool.



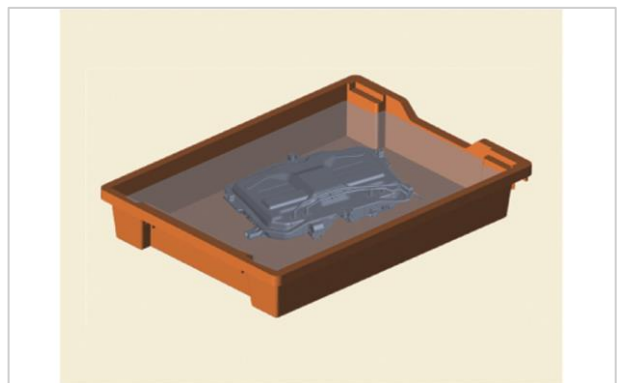
Battery discharging in the water pool

8.3 Damaged Battery Storage

- To store the damaged battery safely, the battery must be discharged.
- If the battery can be removed from the vehicle, discharge the battery to prevent re-ignition.
- Discharge up to 1 volt per a cell. (PHEV: 96cell)
- Prepare water that does not contain salt such as tap water or pond water.
- Leave the battery in water for at least 90 hours.
- Then, put salt in water to make 3.5% salt water.
- Wait for an additional 48 hours in salt water.
- Take out the battery from the container and dry it.

WARNING

- Extinguish all smoke, spark, flame around the vehicle.
- Electrolyte solution is an irritant to skin.
- Do not touch or step on the spilled electrolyte.
- If electrolyte leak occurs, wear appropriate solvent resistant PPE and use soil, sand, or a dry cloth to clean up the spilled electrolyte. Be sure to adequately ventilate the area.



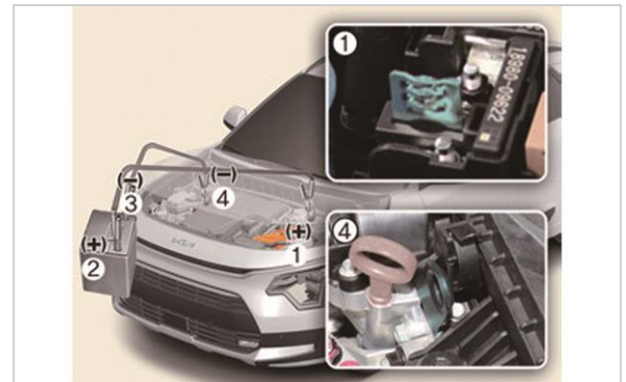
9.1 Emergency Starting

Jump Starting

Do not attempt to jump start the battery pack, high-voltage, as it cannot be jump started. In case of full discharge of the battery pack, high-voltage, the vehicle must be towed as mentioned on the page 28. In case the auxiliary battery, low-voltage is discharged, connect a starting device to the jump terminal in the motor room. Refer to the “Emergency Starting” section of the Owner’s Manual for additional information. Connect jumper cables in the order shown in the image and disconnect in reverse order.

Jump Starting Procedure

1. Position the vehicles close enough that the jumper cables will reach, but do not allow the vehicles to touch. Avoid fans or any moving parts in the engine compartment at all times, even when the vehicles are turned off.
2. Turn off all electrical devices such as radios, lights, air conditioning, etc. Put the vehicles in P (Park) and set the parking brake. Turn both vehicles OFF.
3. Connect the jumper cables in the exact sequence shown in the illustration. First, connect one jumper cable to the charging terminal inside the engine compartment fuse panel (1).



4. Connect the other end of the jumper cable to the red, positive (+) battery/jumper terminal of the assisting vehicle (2).
5. Connect the second jumper cable to the black, negative (-) battery/chassis ground of the assisting vehicle (3).
6. Connect the other end of the second jumper cable to a solid, stationary, metallic point away from the battery (4). Do not allow the jumper cables to contact anything except the correct battery or jumper terminals or the correct ground. Do not lean over the battery when making connections.
7. For hybrid vehicle, press the 12 V BATT RESET button.



8. Then, start the engine of the assisting vehicle and let it run for a few minutes.

9. Start your vehicle as soon as possible. After starting the vehicle (READY indicator on), operate the vehicle safely outdoors in ready mode stopped and/or drive it for 30 minutes total to charge the battery, low-voltage fully.

If your vehicle will not start after a few attempts, it probably requires servicing. In this event please seek qualified assistance. If the cause of your battery discharging is not apparent, you should have your vehicle checked by an authorized Kia dealer.

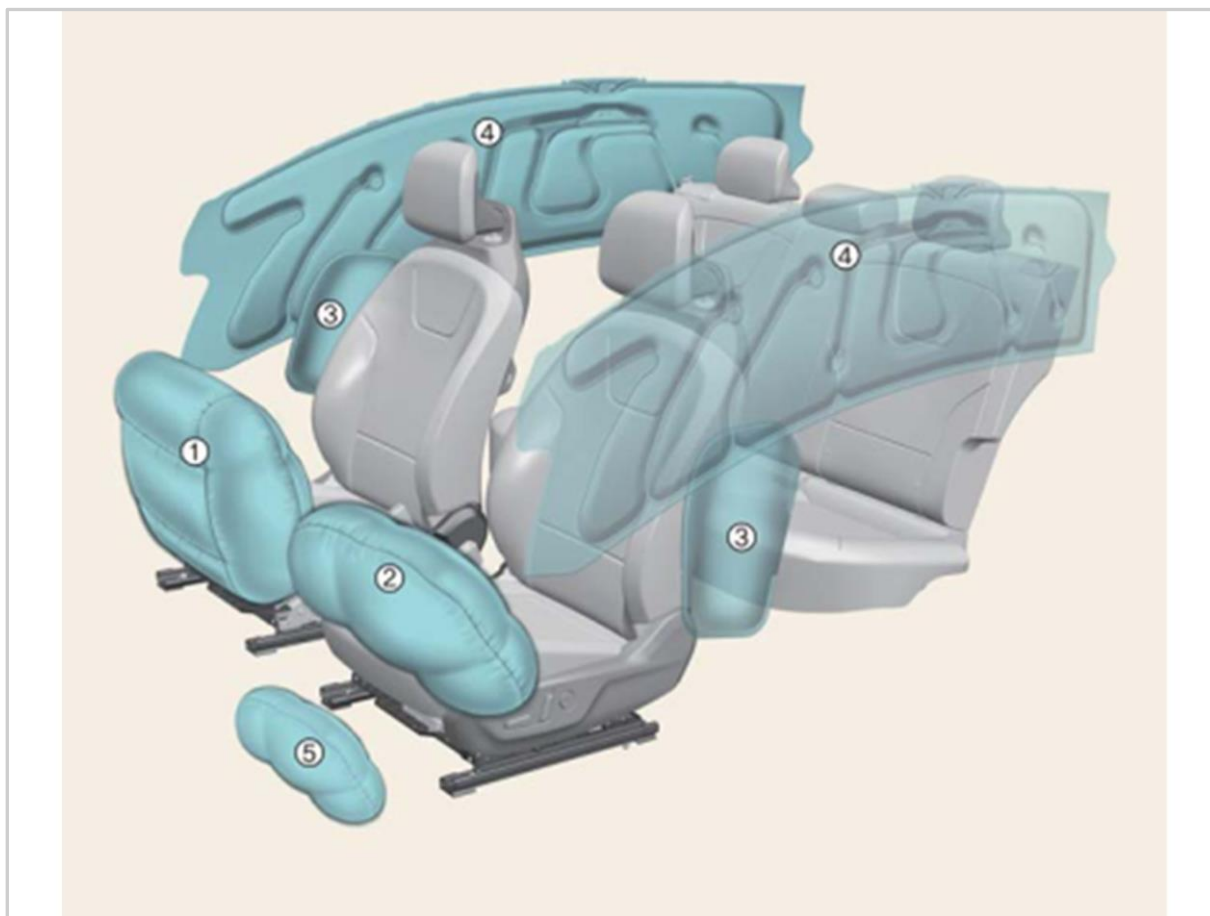


WARNING

Do not attempt to jump start the Niro PHEV battery pack, high-voltage.

Failure to follow these instructions will lead to serious bodily injury or death by electrical shock.

9.2 Airbag system (SRS : Supplemental Restraint System)



Airbag

Note that the actual airbags and seats in the vehicle may differ from the illustration.

Airbags of the Niro PHEV are installed in the areas shown in the image. Before performing emergency procedures, make sure the vehicle ignition switch is turned off and disconnect the negative connector from the auxiliary battery, low-voltage and the service interlock connector to prevent accidental deployment of undeployed airbags.

1. Passenger's front air bag
2. Driver's front air bag
3. Side air bags
4. Curtain air bags
5. Driver's knee air bag

Seat Belt Pretensioner

In the Niro PHEV, the driver's and front passenger's seat belts are equipped with pretensioners.

When the seat belt pretensioners are activated in a collision, a loud noise may be heard and fine dust, which may appear to be smoke, may be visible in the passenger compartment.

These are normal operating conditions and are not hazardous. The seat belt pretensioner assembly mechanisms may become hot during activation, and may need several minutes to cool down after they have been activated.



WARNING

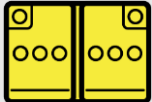
Undeployed airbags

To avoid injury caused by accidental deployment of undeployed airbags.

- Do not cut the red-colored part shown in the image above and left side.
- Make sure the vehicle ignition switch is turned off, disconnect the negative connector from the engine room and wait 3 minutes or longer to allow the system to deactivate.

Failure to follow any of these instructions may result in serious injury or death from accidental deployment of the airbag system.

10. EXPLANATION OF PICTORGRAMS USED

Figure	Meaning
	Plug in hybrid electric vehicle
	High-voltage component
	Use thermal Infrared camera
	General warning sign
	Air-conditioning component
	Air-conditioning line
	Airbag inflator
	Airbag
	SRS control unit
	Battery, low-voltage
	Seat belt pretensioner
	High-voltage power cable
	Battery pack, high-voltage
	High strength zone

[illegible]



Movement that inspires